CHAPTER 14

Volatility and Correlation Structure

14.1 Two functions, and a curse of dimensionality

Chapter 13's drift formula requires, as inputs, an instantaneous volatility function $\sigma_i(t)$ for every forward rate and a correlation ρ_{ij} for every pair. For a realistic tenor structure — quarterly resets out to 30 years is 120 rates — specifying these completely freely means 120 volatility *functions* of time and a 120×120 correlation *matrix*: thousands of free numbers, vastly more than the handful of liquid cap and swaption prices available to calibrate against. This chapter is about the parametric shortcuts the industry has converged on to make this tractable: forms flexible enough to fit the market, rigid enough to be pinned down by a realistic number of observed prices, and economically sensible enough to extrapolate reasonably where no direct quote exists.

14.2 Parameterizing volatility

Piecewise-constant, the simplest workable choice

The simplest sensible choice makes $\sigma_i(t)$ piecewise constant between tenor dates: $\sigma_i(t) = \sigma_i^{(k)}$ for $T_{k-1} \le t < T_k$. This directly matches the market's own flat-per-caplet quoting convention (Chapter 6) and is fully general within that constraint, but it still leaves one free number per rate per time bucket — a lot of parameters, and no guarantee the resulting surface behaves sensibly between calibration nodes.

Rebonato's parametric form

A far more common choice in practice constrains $\sigma_i(t)$ to depend only on the *time to expiry* of rate i, $x = T_i - t$, through a single functional form shared across every rate, with only a small number of free parameters total.

Key result: Rebonato's humped volatility form

$$\sigma_i(t) \;=\; \phi_i \left[(a + b(T_i - t))\, e^{-c(T_i-t)} + d\right]$$

with global parameters a, b, c, d shared by every rate, and an optional per-rate scaling ϕ_i (often fixed near 1, used only to allow small individual adjustments once the global shape is fit). The functional shape — rising from a floor, humping up at a moderate time-to-expiry, then decaying back toward a long-run floor d — is chosen specifically because it reproduces the widely observed empirical fact that the volatility of a forward rate typically *peaks* at a time-to-expiry of roughly one to two years, rather than being monotonic in time.

ON THE DESK

The empirical hump this form captures is real and worth understanding economically, not just fitting mechanically: a very-short-dated rate (days to expiry) tends to have relatively low residual volatility because the market already has a firm, high-confidence view of where it will fix; a rate with roughly a year to go sits in the zone of maximum genuine uncertainty about the medium-term rate path (policy decisions, economic data); a very long-dated rate (10, 20, 30 years out) tends to have lower volatility again because mean-reversion in the economy dampens the plausible range of very-long-run rate outcomes. Fitting a, b, c, d to a cap curve is, in effect, fitting this three-phase story quantitatively.

Because $\sigma_i(t)$ depends only on $T_i - t$ (time to expiry) rather than on calendar time and expiry separately, this form is called **time-homogeneous**: the volatility a rate carries when it has, say, two years left to run is the same function of "two years left" regardless of which specific tenor date T_i it is or when calendar time t currently sits. This is a deliberate, economically motivated constraint — it is what lets a small number of global parameters describe the volatility of every rate in the curve at once, and what makes the fitted model's implied volatility *surface* behave sensibly as time passes and today's forward rates roll down toward expiry.

Worked example 14.1 — reading off a humped shape.
Fit parameters $a = 0.05, b = 0.30, c = 1.20, d = 0.09$ (all dimensionless, times in years) have been calibrated to a cap curve. Evaluate $\sigma_i(t)$ for three rates: one 3 months from expiry, one 1.5 years from expiry, one 15 years from expiry.
Step 1 — $x = 0.25$.

$$\sigma = (0.05 + 0.075) \times e^{-0.30} + 0.09 = 0.125 \times 0.7408 + 0.09$$

$$= 0.0926 + 0.09 = 0.1826 \;\rightarrow\; 18.3\%$$

Step 2 — $x = 1.5$ *(near the hump).*

$$\sigma = (0.05 + 0.45) \times e^{-1.80} + 0.09 = 0.50 \times 0.1653 + 0.09$$

$$= 0.0827 + 0.09 = 0.1727 \;\rightarrow\; 17.3\%$$

Step 3 — $x = 15$ *(far out).*

$$\sigma = [(0.05 + 0.30 \times 15)e^{-1.20\times 15} + 0.09] \approx [4.55 \times (2.5 \times 10^{-8}) + 0.09] \approx 0.0900 \;\rightarrow\; 9.0\%$$

Sanity check. The long-dated value converges to almost exactly $d = 9.0\%$, as the formula's construction guarantees ($e^{-cx} \to 0$ as $x \to \infty$), and the two shorter values are both meaningfully above that floor, consistent with the intended humped shape — though with these particular parameters the 3-month value happens to sit slightly above the 1.5-year value rather than clearly below it, a reminder that the "hump" peak location itself depends on the specific a, b, c combination and is not guaranteed to land exactly at any particular time-to-expiry without checking the actual fitted numbers.
Interpretation. A single set of four numbers, a, b, c, d, has just produced a plausible, smoothly-varying volatility for every rate on the curve at every point in its life — this is the entire value of a parametric form: 4 numbers standing in for what would otherwise be hundreds of independent inputs, at the cost of some flexibility to fit every individual market quote perfectly.

14.3 Parameterizing correlation

The correlation matrix ρ_{ij} needs its own parametric shortcut, for the same reason: 120×120 free correlations is neither estimable from available data nor calibratable from a handful of swaption prices.

A simple parametric form

Key result: a common two-parameter correlation form

$$\rho_{ij} \;=\; \rho_\infty + (1-\rho_\infty)\, e^{-\beta\,|T_i - T_j|}$$

Correlation decays exponentially with the *distance in tenor* between two rates, from 1 (a rate is perfectly correlated with itself, $i = j$) down toward a long-run floor $\rho_\infty \in [0,1)$ as the rates get further apart on the curve, at a rate controlled by β. This single form guarantees, by construction, a valid correlation matrix (symmetric, $\rho_{ii} = 1$, and, given a nonnegative β, positive semi-definite) with only two free parameters.

Economically, this says nearby points on the curve (a 5-year and a 5.25-year forward rate) move almost in lockstep, while distant points (a 2-year and a 25-year rate) move more independently — a pattern strongly supported by historical curve data and essential for getting swaption prices right, since a swap rate's volatility (Chapter 7's unfinished business) depends critically on how correlated its underlying forward rates actually are.

Rank reduction: fewer factors than rates

A full $N \times N$ correlation matrix, however parsimoniously parameterized, still implies an N-factor model — N independent sources of Brownian noise, one per rate. This is both computationally wasteful (simulating 120 correlated Brownian motions when the curve's actual observed movement is dominated by two or three principal shapes — roughly, level, slope, and curvature) and numerically awkward for certain calibration and simulation algorithms. **Rank reduction** replaces the full correlation matrix with a low-rank approximation.

Key result: rank reduction via eigenvalue truncation

Given a full correlation matrix ρ, compute its eigenvalue decomposition $\rho = Q\Lambda Q^\top$, keep only the r largest eigenvalues (typically $r = 3$ to 5 is enough to explain well over 95% of historical curve variance), and rescale the resulting rows to restore unit diagonal. The result is a rank-r correlation matrix, implementable in simulation using only r independent Brownian motions per rate (via an $N \times r$ loading matrix) instead of N — often a 20–40$\times$ reduction in the number of random draws needed per time step, with minimal loss of fit quality.

ON THE DESK

That two or three factors explain most of the curve's historical variance is not a modelling convenience invented for computational reasons — it is a genuine, robust empirical finding from principal component analysis of historical yield curve changes, going back decades across essentially every major government bond and swap market studied. The first principal component is almost always a near-parallel *level* shift; the second is a *slope* (steepening/flattening) move; the third, when it matters, is a *curvature* (butterfly) move. A trader's intuitive language — "the curve bear-steepened today" — is describing exactly this decomposition in plain English, and it is why a 3-factor BGM implementation is a common, well-justified default rather than an arbitrary simplification.

PITFALL

Rank reduction is not free: truncating a correlation matrix's eigenvalues and rescaling to restore a unit diagonal necessarily changes the *off-diagonal* correlations slightly from their original fitted or historical values — the rescaled matrix is the best rank-r approximation in a specific mathematical sense, not an exact reproduction. For products whose value is highly sensitive to the precise correlation between two specific, nearby rates (certain spread and range-accrual structures), checking how much rank reduction has actually distorted the correlations that matter most to that specific product, rather than trusting an aggregate "95% of variance explained" headline number, is worth the extra step.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Explain why a fully free volatility function per rate and a fully free correlation matrix are both computationally and statistically infeasible to calibrate directly, and why a parametric form is needed.
2. State Rebonato's humped volatility form and explain, economically, why volatility is often assumed to peak at a moderate time-to-expiry rather than being monotonic.
3. Explain what rank reduction does to a correlation matrix, why 2–3 factors typically suffice for a rates curve, and what is given up in exchange for the computational saving.